River flow forecasting for water allocation on the Potomac and flood warning at Gandak Barrage

Abstract

Can a river forecast improve on carrying the latest flow measurement forward, and does that improvement change a water management decision? We examined two cases using historical observations. We forecast next-day discharge on the Potomac River near Washington, DC, and hourly inflow at Gandak Barrage in India.

For the Potomac, we used daily records from 1990 to 2025 to compare three long short-term memory networks, a Transformer and a Bayesian ensemble. Across 2,192 test days in 2020 to 2025, the stacked network achieved the lowest root-mean-square error (111.2 m^3/s, compared with 145.8 m^3/s for the last observed flow). In a simulation with hypothetical water demand, basing allocations on the fifth percentile of the ensemble forecast reduced the share of days when allocations exceeded available supply from 63.5% to 1.5%. This more conservative policy also left more demand unmet.

At Gandak Barrage, we fitted ridge regression models to recent inflow measurements to forecast 1 to 72 hours ahead during the 2026 monsoon. Carrying the last reading forward yielded lower mean absolute error at all 72 lead times. On 27 September, the highest inflow forecast over the next three days was 12,519 m^3/s. Observed inflow reached 15,396 m^3/s just 14 hours later. The Potomac case quantifies a trade-off between supply reliability and unmet demand. The missed Gandak crest motivates testing whether upstream gauges and rainfall forecasts can provide earlier warning.